

Fault Diagnosis for Electric Drive Systems of Electrified Vehicles Based on Structural Analysis

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Abstract—Functional safety is of great importance for electric and hybrid electric vehicles (EV/HEVs). One way to improve functional safety of EV/HEVs is to develop reliable and robust onboard diagnostics (OBD) so that, once a component fault is detected, effective remedial actions are taken to avoid system failures. In this paper, we develop a systematic model-based diagnostic approach based on structural analysis for electric drive systems that can form the basis of OBD design for EVs. The structural analysis approach for fault detection and isolation (FDI) evaluates a system's structural model, using the mathematical model of the system in matrix form, from which it is possible to determine the analytic redundancy and to design the structured residuals. In this paper, we demonstrate the application of this methodology by carrying out the design of a diagnostic approach for permanent-magnet synchronous machine (PMSM) drive systems of EVs, with specific focus on sensor fault diagnosis. The diagnostic approach is first verified through a simulation study on the EcoCAR2 vehicle, which is a plug-in HEV developed by the Center for Automotive Research at The Ohio State University, and is then further validated through an experimental study on a test bench consisting of a three-phase inverter enabled by the TMS320F28035 digital signal processor and a PMSM.

Index Terms—Electric vehicles, fault diagnosis, functional safety, permanent-magnet synchronous machine (PMSM), structural analysis.

I. INTRODUCTION

FUNCTIONAL safety is increasingly important for future automobile development, such as for electric and hybrid vehicles (EV/HEVs), due to the growing complexity and the wide employment of electrical and electronic components. To

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ensure functional safety, the automobile industry has developed its own functional safety standard (ISO 26262), which defines the functional safety requirements and life-cycle management for the safety-related components of automobiles in different phases of the development process [1]. One of the effective ways to guarantee functional safety of HEV/EVs is to develop onboard diagnostics (OBD) that are in compliance with the functional safety standards. Cordoba *et al.* [2] studied the potential failure modes and their consequences for the key components, such as electric motors (EMs) and power electronic devices in an electrified powertrain. Faults in any of these electrical or electronic components, for instance, a short or open circuit in one of the motor windings, can lead to serious problems in vehicles, such as degraded performance, increased noise and vibration, unintended torque requests, etc., which could negatively affect the vehicle's functional safety. Therefore, it is extremely important to develop reliable and robust diagnosis and fault-tolerant control for electrified powertrains, to guarantee safe and reliable operation.

Substantial research work has been conducted in developing fault diagnostic strategies for electric drive systems, which are key subsystems of EVs. We cite, for example, work related to power-inverter faults [3]–[5] and EM faults [6]–[9]. The diagnostic approaches described in these references are often data driven, presenting significant challenges with regard to calibration requirements. Sensor faults, on the other hand, can be detected by model-based approaches that compare the measured signals with estimated ones, often using state observers [10], [11]. While observer-based FDI techniques have been shown to be quite effective in detecting incipient faults, isolation of faults may become difficult when multiple sensor faults are considered in the system. With respect to fault diagnosis of electric drive systems for EV applications, we find few references on systematic approaches on performing OBD. For example, Lee *et al.* [12] studied demagnetization fault diagnosis of permanent-magnet synchronous machines (PMSMs) used in EVs, and Jeong *et al.* [13] discussed fault detection for various sensor faults in EVs. However, these references are still about specific faults inside the electric drive system, rather than generalized in a systematic way.

In this paper, we present a systematic fault detection and isolation (FDI) methodology based on structural analysis, which is not specific to any particular type of fault or particular subsystem but is generally applicable to any automotive system. In particular, we focus on the use of this methodology for the diagnosis of electric drive system sensor malfunctions in EV/HEV applications. For control purposes, an electric drive system in an EV is usually embedded with various types of sensors, including current sensors, voltage sensors, speed/position sensors, etc. In some earlier papers, hazard analysis [2] and

plant and fault modeling techniques [14] for defining diagnostic requirements for electrified powertrains have been documented. It has been found out that sensor faults, such as a bias in one of the three phase current sensors or position sensors, can lead the controller to generate inappropriate control actions, resulting in serious consequences such as significant torque oscillations and unstable battery voltages. Sensor faults usually occur in the form of a bias or a gain drift, which are usually difficult to detect if the fault has a small magnitude and the measurements are noisy. Therefore, in this paper, we illustrate the effectiveness of this methodology by considering sensor fault diagnosis in a PMSM drive system, including current sensors, voltage sensors, and position sensors, in particular about incipient faults that are of small magnitude.

Structural analysis for FDI is a model-based methodology that uses the *structural model* of a system to identify the ARRs of the system that allows us to diagnose faults in the system [15], [16]. The advantage of the structural methodology is that it can efficiently analyze the detectability and isolability properties of various faults that could occur in a complex system by looking into the system's structure in graphical or matrix form, without a deep knowledge of the system dynamics or exact mathematical expressions of the system model. It also serves as a guide to the design of diagnostic tests by showing the analytic redundant part of a system, which is required for any diagnosis system design. Therefore, it is applicable in the early stages of design process to any diagnosis system, when the final design of the diagnosis system is not completed. A further advantage of structural analysis approach is that it decomposes a complex system into smaller subsystems; this decomposition allows for efficient design of diagnostic algorithms that are more easily implementable. This approach is particularly useful in diagnosis for large complex systems that have many internal interactions. The structural analysis approach has been well developed in the literature, most of which focus on its theoretical development, e.g., in [17]–[21]. As to the application side, the structural analysis approach has been applied to engine systems [22]–[24], battery systems [25], and hybrid vehicle systems [26], [27], but none of them have been applied to electric drive systems of EVs.

After conducting structural analysis, one still needs to develop specific diagnostic algorithms for implementation, which is a subject that is well understood [15], [28]–[30]. Nevertheless, structural analysis provides guidance as to the architecture and structure of the diagnostic algorithms and helps divide the system into subsystems of manageable complexity. The choice of specific diagnostic algorithms, for example, whether it is based on linear/nonlinear observers, parity equations, or other diagnostic methods, depends on the specifics of the applications and nature of the problem. In this paper, we present the general aspects of this approach, and we show how this methodology can be applied to an automotive electric drives, showing both a general graphical approach for analyzing the analytic redundancy of a system, and the numerical design of residual generators for diagnosing sensor faults in a PMSM drive system. In particular, this paper uses the structural analysis approach to decompose the original electric drive system into smaller subsystems, where the existing diagnostic algorithms such as nonlinear observer techniques can be more easily applied for implementation of the residuals. On the application side, this

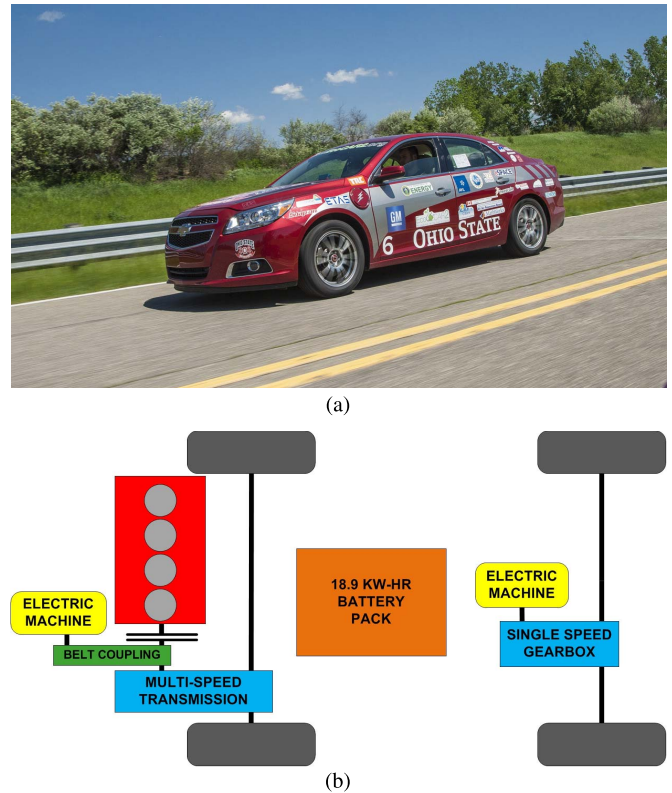


Fig. 1. EcoCAR2 prototype vehicle. (a) EcoCAR2 vehicle. (b) EcoCAR2 vehicle architecture.

paper uses a case study of a prototype vehicle developed by the EcoCAR2 team at The Ohio State University Center for Automotive Research (OSU CAR). EcoCAR2 is a plug in series-parallel HEV, as shown in Fig. 1. The models used in this paper are based on the experimentally validated models that are derived from the EcoCAR2 development process. The diagnostic strategy for the electric drive system is verified through simulations in Matlab/Simulink, using the parameters from the rear powertrain of the EcoCAR2 vehicle. To further validate the diagnostic strategy, we conduct an experimental study on an actual electric drive system consisting of a PMSM, a three-phase inverter, and a digital signal processor (DSP)-based controller from Texas Instruments (TI).

This paper is organized as follows. In Section II, some theoretical background on model-based diagnosis and structural analysis for FDI are introduced. In Section III, we conduct structural analysis for a PMSM drive system, where the structural model of the PMSM drive system is developed and detectability and isolability analysis for the considered sensor faults are performed. Then, in Section IV, we describe diagnostic tests design using minimum structurally overdetermined (MSO) sets for the PMSM drive system and derive residuals for the selected diagnostic tests. Section V shows the results of a simulation study, using parameters from the rear powertrain of the EcoCAR2 vehicle. In Section VI, we discuss experimental results from an electric-drive-system test bench. Section VII applies the cumulative sum control chart (CUSUM) algorithm to the residuals we obtain from experimental results for diagnostic decision. Finally, in Section VIII, we present a summary and conclusion of this paper, as well as our future work.

paper, it is assumed that the electric motor is rigidly connected to the drive shaft so that motor speed is proportional to the wheel speed by the gearbox ratio. We also assume that, from a practical consideration, the measurements are the three phase currents i_a, i_b , and i_c , the three phase voltages V_a, V_b , and V_c , the motor electric angle θ_r , and the wheel speed ω_{wh} .

$$\begin{aligned}
 e_1 : V_d &= \sqrt{\frac{2}{3}} \left[V_a \cos \theta_r + V_b \cos \left(\theta_r - \frac{2}{3}\pi \right) \right. \\
 &\quad \left. + V_c \cos \left(\theta_r + \frac{2}{3}\pi \right) \right] \\
 e_2 : V_q &= \sqrt{\frac{2}{3}} \left[-V_a \sin \theta_r - V_b \sin \left(\theta_r - \frac{2}{3}\pi \right) \right. \\
 &\quad \left. - V_c \sin \left(\theta_r + \frac{2}{3}\pi \right) \right] \\
 e_3 : \frac{di_q}{dt} &= -\frac{R_s}{L_q} i_q - \frac{L_d}{L_q} P \omega_m i_d - \frac{\lambda_m}{L_q} P \omega_m + \frac{V_q}{L_q} \\
 e_4 : \frac{di_d}{dt} &= \frac{L_q}{L_d} P \omega_m i_q - \frac{R_s}{L_d} i_d + \frac{V_d}{L_d} \\
 e_5 : i_a &= \sqrt{\frac{2}{3}} [i_d (\cos \theta_r) - i_q \sin(\theta_r)] \\
 e_6 : i_b &= \sqrt{\frac{2}{3}} \left[i_d \cos \left(\theta_r - \frac{2}{3}\pi \right) - i_q \sin \left(\theta_r - \frac{2}{3}\pi \right) \right] \\
 e_7 : i_c &= \sqrt{\frac{2}{3}} \left[i_d \cos \left(\theta_r + \frac{2}{3}\pi \right) - i_q \sin \left(\theta_r + \frac{2}{3}\pi \right) \right] \\
 e_8 : \frac{d\theta_r}{dt} &= P \omega_m \\
 e_9 : T_e &= 1.5P [\lambda_m i_q + (L_q - L_d) i_d i_q] \\
 e_{10} : \frac{d\omega_m}{dt} &= \frac{1}{J} (T_e - b \omega_m - T_L) \\
 e_{11} : \omega_{wh} &= \frac{\omega_m}{R_{gb}} \\
 e_{12} : V_{veh} &= \omega_{wh} \times R_{wh} \\
 e_{13} : y_{V_a} &= V_a + f_{y_{V_a}} \quad e_{16} : y_{i_a} = i_a + f_{y_{i_a}} \\
 e_{14} : y_{V_b} &= V_b + f_{y_{V_b}} \quad e_{17} : y_{i_b} = i_b + f_{y_{i_b}} \\
 e_{15} : y_{V_c} &= V_c + f_{y_{V_c}} \quad e_{18} : y_{i_c} = i_c + f_{y_{i_c}} \\
 e_{19} : y_{\theta_r} &= \theta_r + f_{y_{\theta_r}} \quad e_{20} : y_{\omega_{wh}} = \omega_{wh} + f_{y_{\omega_{wh}}} \quad (3)
 \end{aligned}$$

2) *Structural Model of PMSM Drive System*: Based on the system's mathematical model, the structural model of the PMSM drive system can be easily obtained as shown in Fig. 3, where the relationships between the equations and variables can be clearly seen. Based on (3), the known variables are the measured outputs from the equations $e_{13} \sim e_{20}$; the remaining variables contained in the model equations are known variables that wait to be calculated from the known variables. The faults considered in this model are all the sensor faults. The set of unknown and known variables and faults are highlighted in Fig. 3.

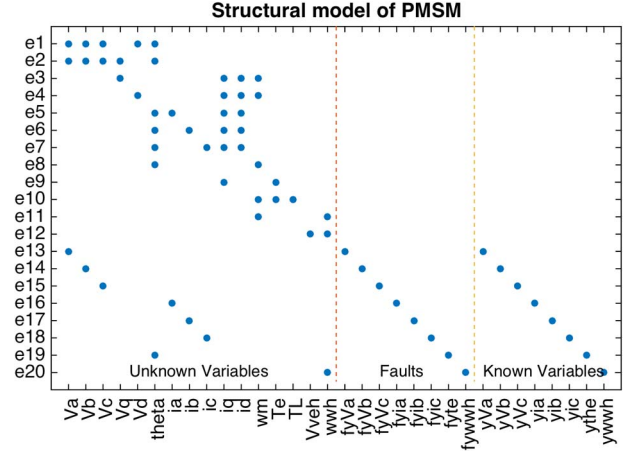


Fig. 3. PMSM drive system structural model.

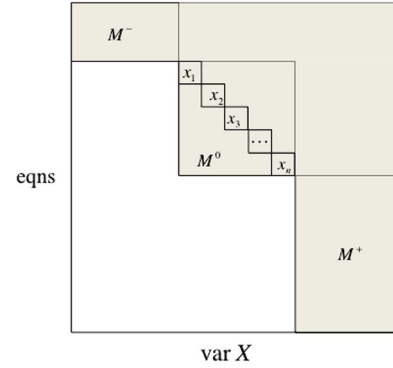


Fig. 4. DM decomposition of a model.

B. Analytic Redundancy of the Model

According to [34], the redundancy of a model can be found through the mathematical tool Dulmage–Mendelsohn (DM) decomposition, which rearranges the rows and columns of a sparse matrix such that it looks like an upper triangular form, imposing the system model to be structured into three parts (which is shown in Fig. 4):

- structurally underdetermined part M^- , where there are fewer equations than unknown variables;
- structurally just-determined part M^0 , where there are equal number of equations as unknown variables;
- structurally overdetermined part M^+ , where there are more equations than unknown variables. This is the part where the analytic redundancy of a model resides.

C. Detectability and Isolability Analysis

Before the appropriate fault detection and isolation algorithms can be applied, it is necessarily to analyze the detectability and isolability of various faults. Krysander and Frisk [19] claimed that a fault can be detected if it lies in the structurally overdetermined part M^+ . For the PMSM drive system, the result of DM decomposition on its structural model is shown in Fig. 5, where the analytic redundant part is denoted in the

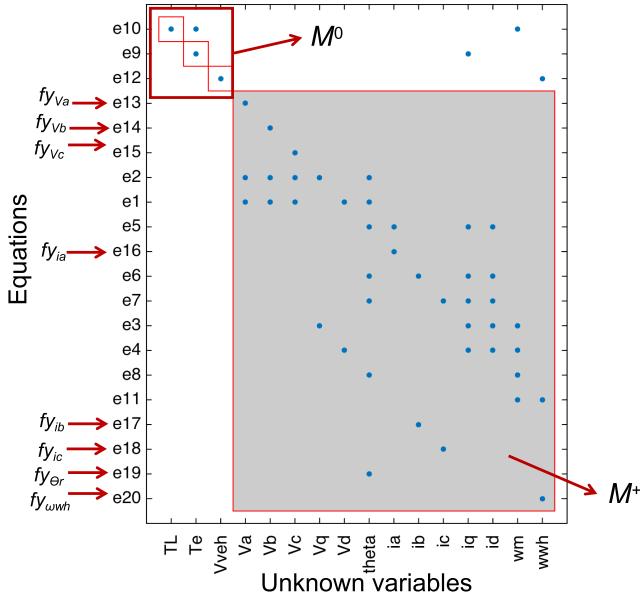


Fig. 5. DM decomposition for the PMSM structural model.

shaded area. As shown in Fig. 5, the considered sensor faults affect equations e_{13} through e_{20} , which lie in the structurally overdetermined part of the structural model (shown in the shaded area in Fig. 5), indicating that all of them can be detected.

According to the definition given by [19]: A fault f_i can be structurally isolated from f_j if the equation that f_i affects lies in the structurally overdetermined part of the model excluding the equation that f_j resides, i.e.,

$$e_{f_i} \in (M/e_{f_j})^+. \quad (4)$$

Based on (4), an isolability analysis can be performed for the sensor faults considered in the model. It can be found that all the faults are isolable from each other.

IV. DIAGNOSTIC TESTS DESIGN

Here, we will introduce the design of diagnostic tests by MSO sets for the considered sensor faults in the PMSM drive system. For each MSO set, a residual is derived to detect specific faults. Multiple MSO sets are selected for fault isolation.

A. Minimum Structurally Overdetermined Sets

According to [16], [18], and [36], a diagnostic test can be formed by utilizing MSO sets, where there is one more equation than unknowns, i.e., with the degree of redundancy of one. MSO sets are designed to use as fewer equations as possible, although they still maintain redundancy to detect specific faults. The use of MSO sets decomposes a complex system into smaller and simpler submodels, from which it is more efficient to design residuals and apply specific diagnostic algorithms. For each MSO set, a consistency relation that is used to form a residual can be found by eliminating the unknown variables sequentially from the associated equations,

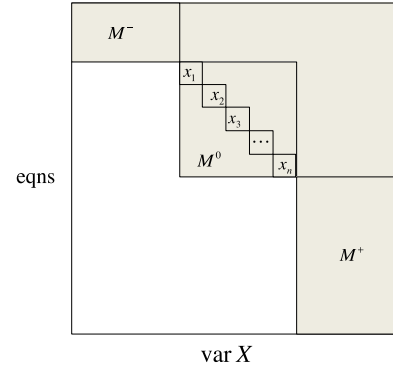


Fig. 6. Diagnosis through MSO sets.

except for the one that is selected to generate a residual. The residuals found in this way are called sequential residual generators (see [16], [17], and [27]). In [20], an efficient algorithm to find all the MSO sets that provide us with all the possible options of diagnostic tests to detect and isolate specific faults is discussed. Whether one chooses to use a MSO and how many MSOs to use depends on the requirements of the diagnostic problem and specifics of the system. A complex model usually contains a combination of algebraic and dynamic equations, which leads its MSOs to possibly include both algebraic and dynamic equations. Therefore, we should consider the integral and derivative causality during selection of MSOs [24], [26], [37].

An MSO set may involve utilizing multiple sensors and faults in any of these sensors can result in a nonzero residual, leading to difficulty in making a diagnostic decision. However, combination of various MSO sets can effectively identify the location of faults since different faults affect these MSO sets in different ways. In conclusion, structural analysis by utilizing MSO sets are useful to detect and isolate faults in an efficient way, with specifics of the problem taken into consideration. The diagnosis system using MSO sets is shown in Fig. 6, where $u_1, u_2, \dots, u_n$ are subsets of system inputs, and $y_1, y_2, \dots, y_n$ are subsets of measurements.

B. Diagnostic Test Selection for PMSM Drives

Following the algorithm described in [16] and [20], a number of MSO sets can be found for the PMSM drive system, as shown in Fig. 7, which enables us to select specific tests to detect and isolate all the sensor faults.

While selecting the MSO sets, one should consider computational complexity and robustness of each MSO set [36]. Some of the MSO sets contain a lot of equations, which may be affected by a large number of faults, thus is not very robust. Therefore, an MSO set that uses as fewer equations as possible to detect certain faults is more preferred. Moreover, while we choose an MSO set, the way in computing each variable should involve as fewer nonlinearities as possible to find unique solutions more easily. For example, as shown from the model equations described by $e_1 \sim e_{20}$, the electric angle variable θ_r appears in coordinate transformations, which are highly nonlinear, making it difficult to determine the angular

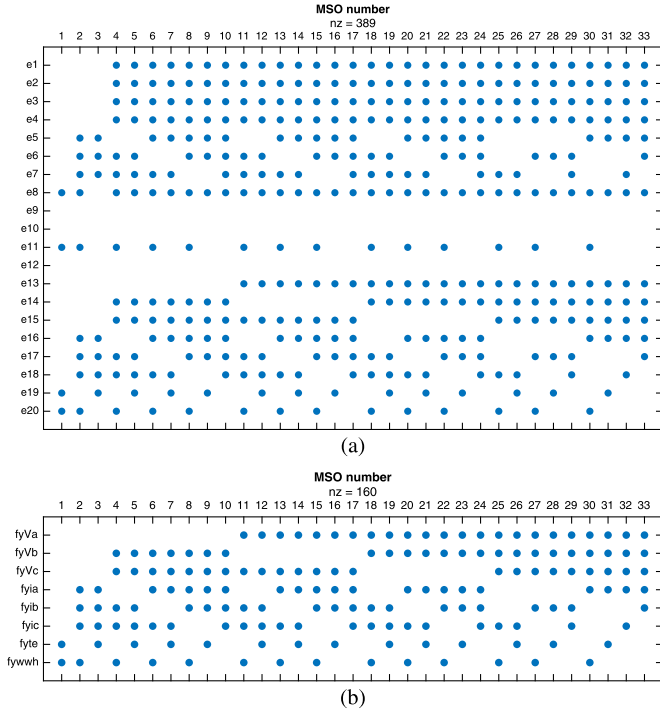


Fig. 7. MSO sets found for PMSM drive system. (a) Equations contained in the found MSO sets. (b) Fault sensitivity of the found MSO sets.

TABLE I
MSO CANDIDATES

	MSO number	Equations contained
MSO1	1	$\{e_8, e_{11}, e_{19}, e_{20}\}$
MSO2	9	$\{e_1 \sim e_6, e_8, e_{14}, e_{15}, e_{16}, e_{17}, e_{19}\}$
MSO3	16	$\{e_1 \sim e_6, e_8, e_{13}, e_{15}, e_{16}, e_{17}, e_{19}\}$
MSO4	23	$\{e_1 \sim e_6, e_8, e_{13}, e_{14}, e_{16}, e_{17}, e_{19}\}$
MSO5	31	$\{e_1, e_2, e_3, e_4, e_5, e_8, e_{13}, e_{14}, e_{15}, e_{16}, e_{19}\}$
MSO6	28	$\{e_1, e_2, e_3, e_4, e_6, e_8, e_{13}, e_{14}, e_{15}, e_{17}, e_{19}\}$
MSO7	26	$\{e_1, e_2, e_3, e_4, e_7, e_8, e_{13}, e_{14}, e_{15}, e_{18}, e_{19}\}$

position from these transformation matrices. On the other hand, it is much simpler and straightforward to calculate θ_r from its measurement equation e_{19} . Therefore, the MSO sets we select should contain e_{19} , in which case we can eliminate a number of useless MSO sets. In terms of isolation properties, one needs to select several MSO sets such that each uses specific sets of measurements and each sensor fault affects different MSO sets. For example, if we wish to identify which of the three current sensor faults is faulty, we could construct three MSOs such that each MSO only uses each of the three current sensors, and it is straightforward to use a residual that contains the variable directly measured by the faulty sensor. Assuming that only one sensor fault occurs at a time, the following seven MSO sets are selected as shown in Table I, where, MSO1 is used to detect wheel speed or motor speed sensor fault, MSO2~MSO4 are used to detect and isolate voltage sensor fault, whereas MSO5~MSO7 are used to detect and isolate current sensor fault. MSO2~MSO7 can be used to detect motor speed/position sensor fault, as well as wheel-speed sensor fault. The combined use of these MSO sets is able to isolate all of the faults considered above, as shown in Table II.

TABLE II
PROPERTIES OF SELECTED MSO SETS

	fV_a	fV_b	fV_c	f_{i_a}	f_{i_b}	f_{i_c}	$f_{\theta}(f_{\omega_m})$	$f_{\omega_{wh}}$
MSO1							X	X
MSO2		X	X	X	X		X	
MSO3	X		X	X	X		X	
MSO4	X	X		X	X		X	
MSO5	X	X	X	X			X	
MSO6	X	X	X		X		X	
MSO7	X	X	X			X	X	

C. Derivation of Sequential Residual Generators

1) *MSO1*: MSO1 uses the simple relationship between the motor speed and wheel speed to test motor speed or wheel-speed sensor faults, the residual can be designed as

$$R_1 = y_{\omega_{wh}} - \frac{\omega_m}{R_{gb}} \quad (5)$$

where $y_{\omega_{wh}}$, ω_m are the measured wheel speed and calculated motor speed, respectively. The motor speed is calculated from the derivative of the measured rotor angular position.

2) *MSO2*: As shown in Table I, this MSO contains voltage equations and voltage measurements in Phases B and C and, therefore, can be used to detect phase B or phase C voltage sensor faults.

a) *Derivation of residuals*: Theoretically, each of the equations contained in this MSO can be selected as the consistency relation and generate a residual. Nevertheless, since we mainly use this MSO to test voltage sensor faults, it is natural to choose an equation that is directly related to the corresponding voltage variables for better detection results. If we eliminate V_a from e_1 and e_2 , we obtain an ARR that is only related to V_d , V_q , V_b , and V_c , i.e.,

$$V_d \sin \theta_r + V_q \cos \theta_r - \frac{\sqrt{2}}{2} V_b + \frac{\sqrt{2}}{2} V_c = 0. \quad (6)$$

Since V_b and V_c are measured, the magnitude of sensor faults will be reflected directly on this ARR; thus, this equation can be used to form a residual to test voltage sensor faults in Phases B and C.

Since i_a and i_b are measured, we can calculate i_q and i_d from e_5 and e_6 as follows:

$$y_{i_q} = -\sqrt{2} \left[y_{i_a} \cos \left(y_{\theta_r} - \frac{2}{3} \pi \right) - y_{i_b} \cos (y_{\theta_r}) \right] \quad (7)$$

$$y_{i_d} = -\sqrt{2} \left[y_{i_a} \sin \left(y_{\theta_r} - \frac{2}{3} \pi \right) - y_{i_b} \sin (y_{\theta_r}) \right]. \quad (8)$$

From e_8 , the motor speed can be calculated from derivation of rotor angular position. Then, from e_3 and e_4 , the input voltages V_d , V_q can be calculated as

$$\begin{aligned} V_q &= L_q \dot{i}_q + R_s i_q + P \omega_m L_d i_d + P \omega_m \lambda_m \\ V_d &= L_d \dot{i}_d + R_s i_d - P \omega_m L_q i_q. \end{aligned} \quad (9)$$

Therefore, a residual can be designed as

$$R_2 = V_d \sin \theta_r + V_q \cos \theta_r - \frac{\sqrt{2}}{2} y_{V_b} + \frac{\sqrt{2}}{2} y_{V_c}. \quad (10)$$

To calculate V_q and V_d , (9) involves derivation of current measurements, which can bring significant noise since electrical signals are of very high frequency. To avoid derivative causality, we can take advantage of existing diagnostic techniques such as sliding-mode observers described in [38], [39]. A sliding-mode observer for the input voltage estimation can be designed as

$$\begin{aligned}\frac{d\hat{i}_q}{dt} &= \frac{1}{L_q}\eta_q - \frac{R_s}{L_q}\hat{i}_q - \frac{L_d}{L_q}P\omega_m\hat{i}_d - \frac{\lambda_m}{L_q}P\omega_m \\ \frac{d\hat{i}_d}{dt} &= \frac{1}{L_d}\eta_d - \frac{R_s}{L_d}\hat{i}_d + \frac{L_d}{L_q}P\omega_m\hat{i}_q\end{aligned}\quad (11)$$

where $\eta_q = K \text{sign}(i_q - \hat{i}_q)$, and $\eta_d = K \text{sign}(i_d - \hat{i}_d)$. The error dynamics is given by

$$\begin{aligned}\dot{e}_{i_q} &= \frac{1}{L_q}V_q - \frac{1}{L_q}\eta_q - \frac{R_s}{L_q}e_{i_q} - P\omega_m\frac{L_d}{L_q}e_{i_d} \\ \dot{e}_{i_d} &= \frac{1}{L_d}V_d - \frac{1}{L_d}\eta_d - \frac{R_s}{L_d}e_{i_d} + P\omega_m\frac{L_q}{L_d}e_{i_q}.\end{aligned}\quad (12)$$

It can be proved that if $K > \max(|V_q|, |V_d|)$, $e_{i_q} = e_{i_d} = 0$ is enforced. The equivalent control is then derived as

$$(\eta_q)_{eq} = V_q, \quad (\eta_d)_{eq} = V_d. \quad (13)$$

To extract the values of V_q , V_d , a first-order filter can be designed as

$$\tau\dot{\hat{V}}_q + \hat{V}_q = \eta_q, \quad \tau\dot{\hat{V}}_d + \hat{V}_d = \eta_d \quad (14)$$

where τ is the time constant of the filter. Then

$$\lim_{t \rightarrow \infty} \hat{V}_q = (\eta_q)_{eq} = V_q, \quad \lim_{t \rightarrow \infty} \hat{V}_d = (\eta_d)_{eq} = V_d. \quad (15)$$

3) *MSO3 and MSO4*: Similar to MSO2, MSO3 relies on the voltage measurements in phase A and phase C, whereas MSO4 uses voltage measurements in phase A and phase B. Therefore, the combination of MSO2, MSO3, and MSO4 can be used to isolate the particular voltage sensor fault. The residuals from MSO3 and MSO4 can be expressed as follows:

$$\begin{aligned}R_3 &= V_d \sin\left(\theta_r - \frac{2}{3}\pi\right) + V_q \cos\left(\theta_r - \frac{2}{3}\pi\right) \\ &\quad + \frac{\sqrt{2}}{2}y_{V_a} - \frac{\sqrt{2}}{2}y_{V_c}\end{aligned}\quad (16)$$

$$\begin{aligned}R_4 &= V_d \sin\left(\theta_r + \frac{2}{3}\pi\right) + V_q \cos\left(\theta_r + \frac{2}{3}\pi\right) \\ &\quad - \frac{\sqrt{2}}{2}y_{V_a} + \frac{\sqrt{2}}{2}y_{V_b}.\end{aligned}\quad (17)$$

4) *MSO5, MSO6, MSO7*: In these three diagnostic tests, motor speed, angular position and voltage information in three phases are all available, whereas each diagnostic tests only contains current information in a single phase (either i_a , i_b , or i_c , respectively); therefore, these three MSOs are capable of isolating the faulty current sensor. We can first use e_3 and e_4 to calculate i_q and i_d ; then, the three phase currents can be obtained by utilizing inverse Park's transformation. The relative

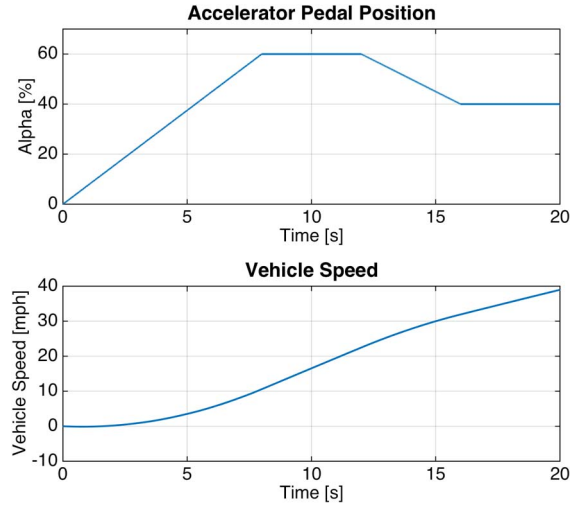


Fig. 8. Acceleration pedal position and vehicle speed.

residuals on the phase currents are given by the difference between the measured currents with the estimated ones, as given by

$$R_5 = y_{i_a} - \sqrt{\frac{2}{3}}[i_d \cos(\theta_r) - i_q \sin(\theta_r)] \quad (18)$$

$$R_6 = y_{i_b} - \sqrt{\frac{2}{3}}\left[i_d \cos\left(\theta_r - \frac{2}{3}\pi\right) - i_q \sin\left(\theta_r - \frac{2}{3}\pi\right)\right] \quad (19)$$

$$R_7 = y_{i_c} - \sqrt{\frac{2}{3}}\left[i_d \cos\left(\theta_r + \frac{2}{3}\pi\right) - i_q \sin\left(\theta_r + \frac{2}{3}\pi\right)\right]. \quad (20)$$

Taking MSO5 as an example, V_d and V_q can be determined from e_1 and e_2 , ω_m is determined from e_8 ; then, i_q and i_d are calculated from e_3 and e_4 , and e_5 becomes a consistency relation to form a residual. The expressions of these residuals indicate that a faulty position sensor and a faulty voltage sensor would also result in nonzero residuals for all the three MSO sets.

V. SIMULATION RESULTS

Here, some simulation results in Matlab/Simulink of the residual responses under faulty conditions for each of the sensor faults are shown. The simulation is based on a PMSM drive system using parameters of the rear powertrain from the OSU EcoCAR2 prototype vehicle. To test the residual responses under variable operating conditions, we made the driver accelerator command variable during the test time window. The accelerator pedal position and the developed vehicle speed is shown in Fig. 8. The road load is also made variable because vehicles usually encounter unexpected road conditions during real maneuvers.

A. Residual Outputs From the Selected MSO Sets Under Faults

In the simulation, the sensor faults are injected at different time intervals: At $t = 3 \sim 3.5$ s, phase A current sensor bias fault is +10 A; at $t = 5 \sim 5.5$ s, phase B current sensor bias

fault is +10 A; at $t = 6 \sim 7$ s, rotor angular position sensor amplitude imbalance fault $\alpha = 0.01$; at $t = 7 \sim 7.5$ s, phase C current sensor bias fault is +10 A; at $t = 10 \sim 11$ s, wheel-speed sensor fault is +2 rad/s bias; at $t = 12 \sim 12.5$ s, phase A voltage sensor fault has +10% gain change; at $t = 14 \sim 14.5$ s, phase B voltage sensor fault has +10% gain change; and at $t = 16 \sim 16.5$ s, phase C voltage sensor fault of +10% gain change. The residual responses are shown in Fig. 9(a)–(c).

As shown in Fig. 9(a), the wheel speed sensor and motor position sensor will result in nonzero residuals in MSO1; however, since the remaining MSOs only rely on the motor speed/angular position measurements, the wheel speed sensor will have no effect on their residuals. In this way, it is possible to isolate wheel-speed sensor fault from motor position sensor fault; voltage sensor faults lead to nonzero residuals in MSO2~MSO4, as shown in Fig. 9(b). However, each of the three phase voltage sensors affect different MSOs, enabling us to isolate the particular faulty voltage sensor. Current residuals are shown in Fig. 9(c), where each of the three phase current sensors only affects a single MSO set, therefore allowing the possibility of identifying which of the current sensors is faulty by comparing the residual response of MSO5~MSO7.

VI. EXPERIMENTAL RESULTS

Here, experimental results on a prototype permanent-magnet synchronous motor are presented to further validate the proposed FDI strategy. Fig. 10 shows the configuration of the experimental platform, which consists of a control board based on DSP TMS320F28035 from TI, a three-phase inverter, and a permanent-magnet synchronous motor with a rated speed of 4000 r/min and rated torque of $0.275 \text{ N} \cdot \text{m}$. The detailed motor specification is shown in Table III. An encoder is installed inside the motor shaft to measure its angular position. A flywheel is connected to the motor shaft to provide external load. The diagnostic algorithm based on sequential residual generators is implemented in Matlab/Simulink, and is loaded into the DSP controller through TI C2000 Support from Simulink Embedded Coder and the TI Code Composer Studio software.

In the experimental study, we consider faults in the two phase current sensors and two phase voltage sensors (phases A and B) and position sensor (encoder) faults, due to the fact that, in a real controller, only two phase current signals are needed for control feedback. Therefore, we choose MSO2, MSO3, MSO5, and MSO6 to detect and isolate the considered sensor faults. Fig. 11 shows the experimental results of the residual responses under various faulty conditions, where in Fig. 11(a) and (b), a current sensor bias fault of 0.6 A is injected at 0.5 s in phases A and B, respectively; in Fig. 11(c) and (d), a 2.5-V bias in the voltage sensor of phase A or phase B is introduced at 0.5 s; and in Fig. 11(e), a position sensor bias fault of 5% is applied at 0.5 s.

As shown in Fig. 11(a) and (b), when there is a bias fault in the phase A current sensor, the current residual from MSO5 will be biased compared with the healthy case, whereas the Phase B current residual from MSO6 stays around zero and *vice versa* for Phase B current sensor fault. Therefore, it is possible to detect and isolate the faulty current sensor; when there is

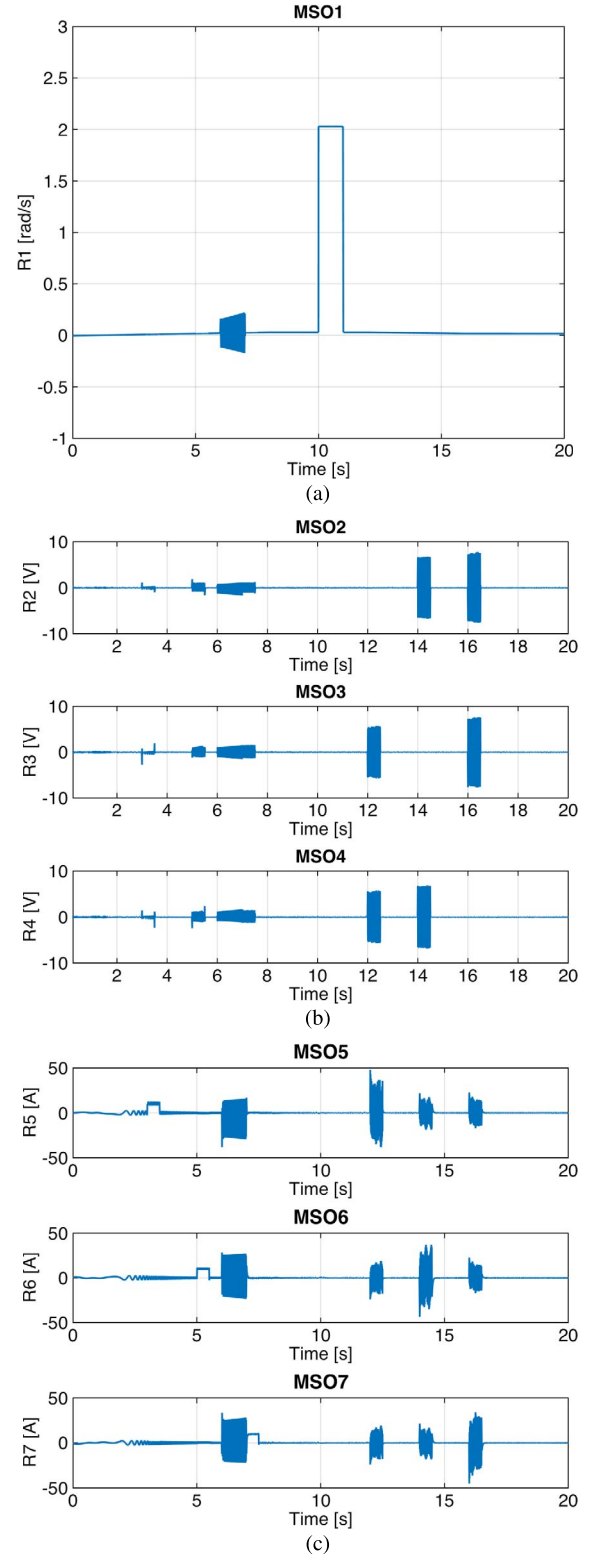


Fig. 9. Simulation results of residuals from MSO sets. (a) Residual from MSO1. (b) Residuals from MSO2, MSO3, and MSO4. (c) Residuals from MSO5, MSO6, and MSO7.

a current sensor bias fault, either in Phase A or in Phase B, both voltage residuals from MSO2 and MSO3 will be biased with greater magnitude of oscillations than in the healthy case. This is due to the fact that when there is a current sensor bias

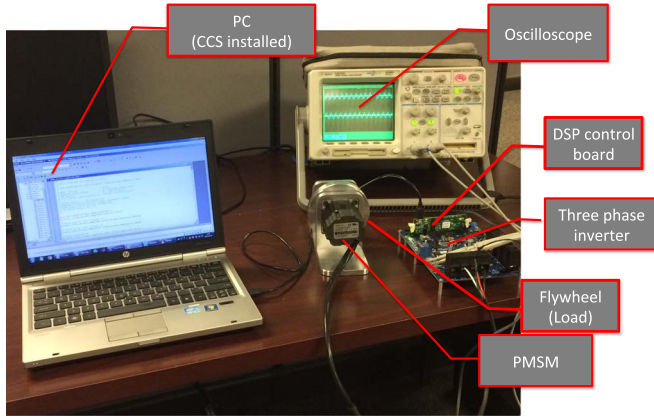


Fig. 10. Experimental setup of an electric drive system.

 TABLE III
MOTOR SPECIFICATIONS

Resistance, phase to phase [Ω]	0.72
Inductance, phase to phase [mH]	0.40
Back EMF constant [V/kRPM]	4.64
Rated torque [Nm]	0.275
Number of poles	8

fault, the residual will have additional sinusoidal terms due to the estimation error in the voltages V_q and V_d .

Similarly, when there is bias fault in phase A (or phase B) voltage sensor, as shown in Fig. 11(c) and (d), the voltage residuals from MSO3 (or MSO2) will be biased, thus enabling us to identify the faulty voltage sensor; when a bias fault occurs in the voltage sensors, both the current residuals in Phase A and Phase B become biased with greater magnitude of oscillations. This is because in MSO5 and MSO6, currents i_q and i_d are estimated based on estimated V_q and V_d , if any of the phase voltage measurements become biased, the estimated V_q and V_d will become oscillating due to coordinate transformation, leading to oscillatory estimated currents.

When there is a position sensor fault, both the estimated currents from voltage measurements and the estimated voltages from current measurements would be incorrect because the rotor angular position signal is used in every coordinate transformation, resulting in nonzero residuals in all of the MSOs, as shown in Fig. 11(e).

VII. DIAGNOSTIC DECISION

A. CUSUM Algorithm

After obtaining the residuals, here, we apply the CUSUM algorithm to the residuals to ensure diagnostic decision [40]. The CUSUM algorithm is based on repeated use of sequential probability ratio test (SPRT), which takes the sum of the log likelihood ratio s_k of an observation from z_1 to z_k

$$S_1^k = \sum_{i=1}^k s_i \quad (21)$$

$$s_i = \ln \frac{p(z_i|H_1)}{p(z_i|H_0)} \quad (22)$$

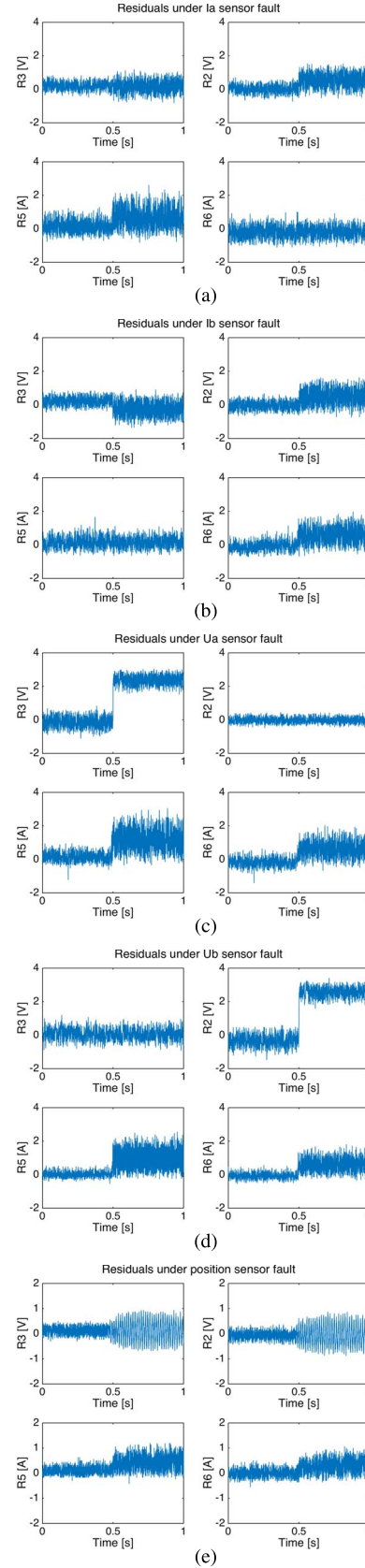


Fig. 11. Experimental results of residual responses from the MSO sets under various sensor faults. (a) Residuals under a 0.6-A bias fault of phase A current sensor at 0.5 s. (b) Residuals under a 0.6-A bias fault of phase B current sensor at 0.5 s. (c) Residuals under a 2.2-V bias fault of phase A voltage sensor at 0.5 s. (d) Residuals under a 2.2-V bias fault of phase B voltage sensor at 0.5 s. (e) Residuals under a 5% bias fault of position sensor at 0.5 s.

where $p(z_i|H_0)$ denotes the probability of an observation z_i under the nonfaulty hypothesis H_0 , and $p(z_i|H_1)$ represents the probability of the observation z_i under the faulty hypothesis H_1 . The decision rule is defined as

$$d = \begin{cases} 0, & \text{if } S_1^k \leq -\epsilon \\ 1, & \text{if } S_1^k \geq \xi \end{cases} \quad (23)$$

where $\epsilon \geq 0$ and $\xi \geq 0$ are selected thresholds. According to [41]–[43], the optimal lower threshold ϵ should be zero.

The idea of the CUSUM algorithm is to restart the SPRT algorithm as long as the previously taken decision is $d = 0$, i.e.,

$$S_k = \begin{cases} S_{k-1} + \ln \left(\frac{p(z_k|H_1)}{p(z_k|H_0)} \right), & \text{if } S_{k-1} + \ln \left(\frac{p(z_k|H_1)}{p(z_k|H_0)} \right) \geq 0 \\ 0, & \text{if } S_{k-1} + \ln \left(\frac{p(z_k|H_1)}{p(z_k|H_0)} \right) < 0. \end{cases} \quad (24)$$

The diagnostic decision logic is defined as

$$\begin{cases} H_0 \text{ is accepted if } S_k < h, \text{ Fault Flag is off} \\ H_1 \text{ is accepted if } S_k \geq h, \text{ Fault Flag is on} \end{cases}$$

where $h > 0$ is the predefined threshold.

B. CUSUM Tests for the Residuals From Experimental Results

As shown in Fig. 11, residual signals from experimental data are very noisy, and the residuals under faults contain significant oscillations, making it impossible to set a constant threshold for diagnostic decision. Therefore, we apply CUSUM algorithms to the residuals based on (24). Before CUSUM algorithms are applied, it is useful to know the probability density functions (PDFs) of the residuals under both healthy and various faulty conditions. Fig. 12 shows the pdfs of residuals under various faults based on the experimental data. As shown in Fig. 12(a) and (b), when there is a bias fault in the current sensor I_a or I_b , the corresponding pdf of the current residual from MSO5 (or MSO6) will be biased by its fault magnitude, whereas the pdfs of the residual from MSO6 (or MSO5) stays almost the same, of which the mean stays around zero. The pdfs of voltage residuals from MSO3 and MSO2 are both biased with significant greater variance compared with those in the healthy case. Similarly, if there is a bias fault in the voltage sensors, the pdf of the corresponding voltage residual will be biased, whereas the current residual pdfs will be biased with greater variance. Finally, when there is a position sensor fault, all of the residuals from the MSO set outputs will have greater variance.

Based on the pdfs under healthy and faulty conditions, we apply CUSUM algorithms to the residuals under various faults, and the results are presented in Fig. 13. As can be seen, by the time a related fault (a fault to which the MSO set is sensitive) occurs, the value of S_k starts to accumulate rapidly; by then, it is possible to set a constant threshold for accurate diagnostic decision.

Finally, the fault flags under various faults can be derived from Table II and are shown in Fig. 14. As shown, the CUSUM algorithm is effective in detecting a fault in the corresponding sensor based on the residual responses.

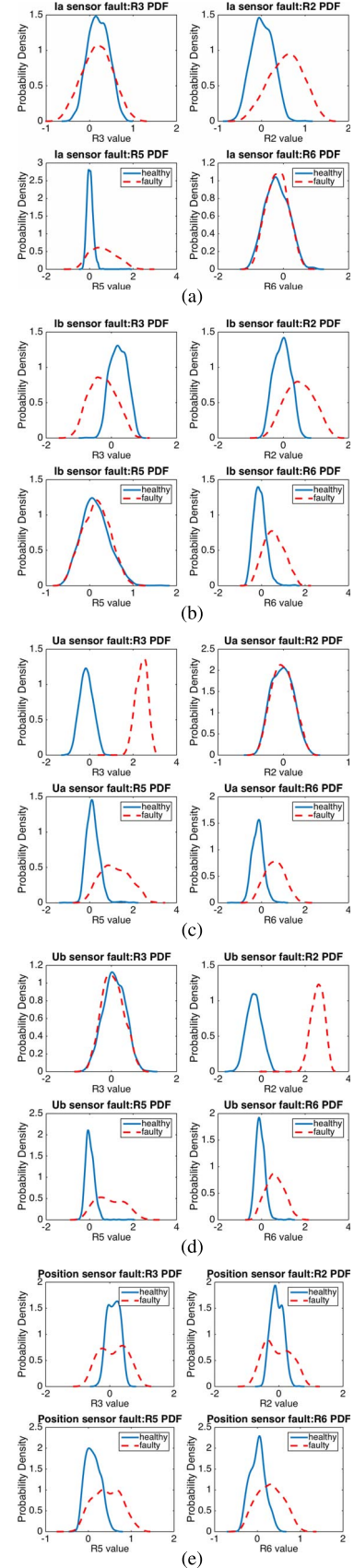


Fig. 12. PDFs of residuals from MSO sets. (a) PDFs of residuals under phase A current sensor fault. (b) PDFs of residuals under phase B current sensor fault. (c) PDFs of residuals under phase A voltage sensor fault. (d) PDFs of residuals under phase B voltage sensor fault. (e) PDFs of residuals under position sensor fault.

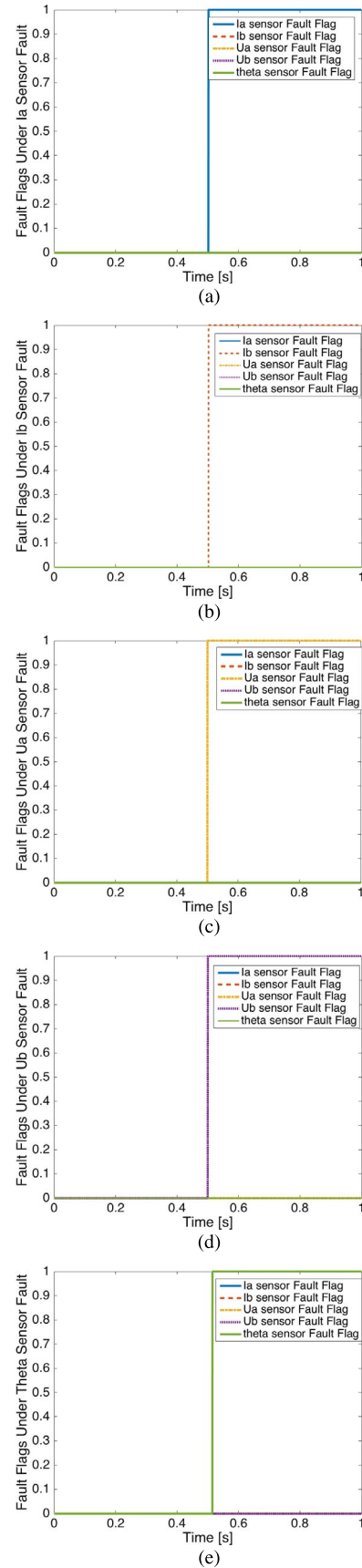
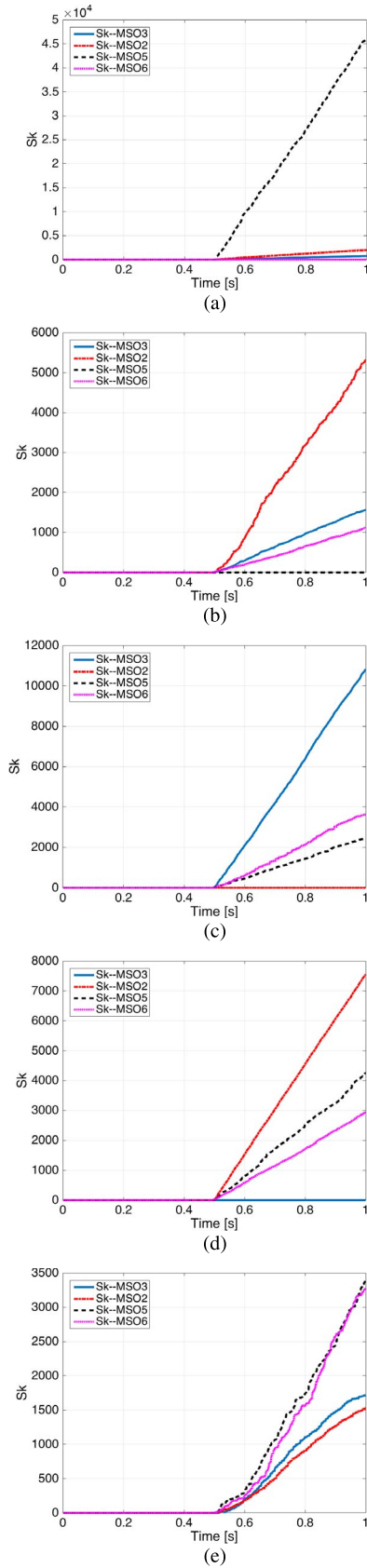


Fig. 13. CUSUM test results for residuals under faults. (a) CUSUM test for residuals under phase A current sensor fault. (b) CUSUM test for residuals under phase B current sensor fault. (c) CUSUM test for residuals under phase A voltage sensor fault. (d) CUSUM test for residuals under phase B voltage sensor fault. (e) CUSUM test for residuals under position sensor fault.

Fig. 14. Fault flags under various sensor faults. (a) Fault flags under phase A current sensor fault. (b) Fault flags under phase B current sensor fault. (c) Fault flags under phase A voltage sensor fault. (d) Fault flags under phase B voltage sensor fault. (e) Fault flags under position sensor fault.

VIII. CONCLUSION

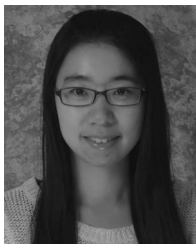
This paper has illustrated a systematic approach based on structural analysis for fault diagnosis of EV drive systems, which can assist the design process of automotive OBD to ensure functional safety. Structural analysis analyzes a system model by structural incidence matrix, from which it is possible to identify the subset of the model where the analytic redundancy resides and identify the detectability and isolability properties for various faults. Structural analysis for FDI gives guidance to systematically design of diagnostic tests and structured residuals for FDI. Various faults can be effectively detected and isolated utilizing a set of MSO sets, each of which contains a subset of model equations and is sensitive to certain faults. Following the structural analysis step, we apply specific diagnostic algorithms to generate residuals and to implement diagnostic tests. The use of MSO sets decomposes the original complex system into smaller subsystems to implement the diagnostic algorithms more easily for each subsystem. This paper has demonstrated the effectiveness of this approach using a case study of fault diagnosis for an electric drive system in a prototype plug-in HEV, i.e., the OSU EcoCAR2 vehicle.

The structural analysis approach described in this paper provides a general framework to analyze the fault diagnosability properties of a system and for designing diagnostic tests. The usefulness of this methodology for FDI is demonstrated by, but not limited to, a special application of sensor faults in an electric drive system, including wheel-speed sensor fault, three phase voltage sensor faults, three phase current sensor faults, position sensor faults. The proposed methodology is validated through both a simulation study conducted in Matlab/Simulink, and an experimental study using a real electric drive system test bench comprised of a PMSM and a DSP-based controller. The simulation results, combined with experimental results, prove the effectiveness the proposed methodology. In the future, we will conduct further validation of the proposed strategy in the EcoCAR2 vehicle.

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